

3. Project Design Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

System Design

The system is designed to control access to records using an Access Control List (ACL). The ACL uses a script to check a specific field value in the record before allowing access.

The basic design flow is:

User → Record Request → ACL → Field Value Check → Access Decision

The ACL acts as a security layer between the user and the requested record.

ACL Design

An ACL is created for the required table and access operation. The ACL contains the required condition or script that determines whether the user can access the record.

The ACL design consists of:

- Target table
- Access operation
- Field value
- Script condition
- Access decision

The script evaluates the required field and returns the appropriate result.

Field-Based Access Design

The main feature of the project is restricting access based on a field value.

For example, consider a Department field.

If the user's department matches the department stored in the record, access is allowed.

If the values do not match, access is restricted.

Example:

Record Department = User Department → Allow

Record Department ≠ User Department → Deny

This design provides record-level access control.

Access Control Workflow

The system follows these steps:

1. A user requests access to a record.
2. The ACL is triggered.
3. The script reads the required field value.
4. The value is compared with the defined condition.
5. The ACL produces an access decision.
6. The user is either allowed or restricted from accessing the record.
7. Security Design

Security is an important part of the system design. The ACL ensures that users cannot access records that do not satisfy the defined condition.

The design reduces unnecessary access to sensitive records and provides an additional layer of record-level security.

System Components

The main components of the proposed system are:

- User: Requests access to a record.
- Table: Stores the application records.
- Field: Contains the value used for access control.
- ACL: Controls access to the record.
- Script: Evaluates the field value.
- Access Decision: Allows or restricts access.
- 8. Expected Design Output

The final design should provide a working access-control flow in which the ACL automatically evaluates the selected field value.

When the condition is satisfied, the record can be accessed. When the condition is not satisfied, the record remains restricted.

Conclusion

The Project Design Phase defines the structure and workflow of the Script-Controlled ACL. The design focuses on using a field value to make record-level access decisions.

The designed workflow provides a clear foundation for the next phase, Project Planning and Development, where the ACL configuration and script will be implemented.